Math 55 Second Midterm Exam, Prof. Srivastava
November 8, 2022, 5:10pm–6:30pm, 2050 VLSB.

Name: ____

SID: ____

GSI: ____

Name of the student to your left: ____________________________
Name of the student to your right: ____________________________

Instructions: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. **Do not unstaple the exam.** Write your name and SID on every page. Show your work — numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.
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Sign here: ____

Question	Points
1	10
2	7
3	7
4	7
5	9
6	10
Total:	50

Do not turn over this page until your instructor tells you to do so.

Name and SID: ______________________________

1. (10 points) Circle true (**T**) or false (**F**) for each of the following. There is no need to provide an explanation. All graph theory definitions are provided on the last page of the exam.

   (a) If $n \geq 3$ and $1 < k < n$ then $\binom{n}{k}$ must be divisible by n. **T** **F**

   (b) if m, n are positive integers and $m \leq n$ then for every $k \leq m$:
   $$\binom{m}{k} \leq \binom{n}{k}.$$
   T **F**

   (c) If a simple connected graph has an Eulerian circuit then it must be $3-$colorable.
   . **T** **F**

   (d) If a simple graph has an Eulerian circuit then it must contain at least two distinct spanning trees. **T** **F**

[Scratch Space Below]

2. (7 points) Prove by induction that 5 divides $n^5 - n$ whenever n is a positive integer.

3. (7 points) Consider the sequence of positive real numbers recursively defined by:

$$x_1 = 1, x_2 = 2, x_{n+1} = \sqrt{1 + x_n + x_{n-1}}.$$

Show using induction that $x_n \leq 4$ for all $n \geq 1$.

4. (7 points) (a) How many subsets of $\{1, 2, \ldots, 17\}$ of size 5 contain at most one even integer? You can express your answer in terms of factorials (no need to simplify).

(b) How many subsets of $\{1, 2, \ldots, 17\}$ of size 5 contain at least one even integer?

5. Recall that the degree sequence of a graph is a list of the degrees of its vertices in nondecreasing order. Give examples of graphs with the following properties, or prove that no such graph can exist.

   (a) (3 points) A simple graph with degree sequence $6, 5, 4, 3, 2, 1$.

   (b) (3 points) A $2-$colorable simple graph with degree sequence $4, 2, 2, 1, 1$.

   (c) (3 points) A tree with degree sequence $3, 3, 2, 2, 1, 1$.

6. (a) (3 points) Prove that if $n \geq 2$ and x, y are positive integers such that $x + y = n$ then
$$\binom{x-1}{2} + \binom{y-1}{2} \leq \binom{n-1}{2}.$$

(b) (7 points) Prove that if G is a simple graph with $n \geq 1$ vertices and at least $\binom{n-1}{2}+1$ edges, then G must be connected.

Name and SID: ______________________________

[Scratch Paper 1]

Name and SID: ______________________________

[Scratch Paper 2]

Graph Theory Definitions

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a finite multiset of $2-$element subsets of V, called **edges**. If E has repeated elements G is called a **multigraph**, otherwise it is called a **simple graph**. Two vertices $x, y \in V$ are **adjacent** if $\{x, y\} \in E$. An edge $e = \{x, y\}$ is said to be **incident with** x and y. The **degree** of a vertex x is the number of edges incident with it.

A **path** of length k in G is an alternating sequence of vertices and edges

$$x_0, e_1, x_1, e_2, \ldots, x_{k-1}, e_k, x_k$$

where $e_i = \{x_{i-1}, x_i\} \in E$ for all $i = 1 \ldots k$. The path is said to **connect** x_0, x_k, **visit** the vertices $x_0, \ldots, x_k$, and **traverse** the edges $e_1, \ldots, e_k$. A path is called **simple** if it contains no repeated vertices. If $x_0 = x_k$ then the path is called a **circuit**. A graph is **connected** if it contains a path between every pair of distinct vertices.

A **k-coloring** of G is a function $f : V \to \{1, \ldots, k\}$ such that $f(x) \neq f(y)$ whenever x is adjacent to y. A graph is $k-$**colorable** if it has a k-coloring.

A graph $H = (W, F)$ is a **subgraph** of G if $W \subseteq V$ and $F \subseteq E$. If $W = V$ then H is called a **spanning** subgraph of G. A **connected component** of G is a maximal connected subgraph of G, i.e. a subgraph $H \subseteq G$ such that H is connected and adding any vertices or edges to H would make it disconnected.

A **cycle** is a circuit with no repeated edges and no repeated vertices, except the endpoints. A graph is called **acyclic** if it does not contain a cycle as a subgraph. A connected acyclic graph is called a **tree**.

An **Eulerian circuit** of a connected multigraph G is a circuit which traverses every edge of G exactly once.